Barangay Administrator
Authorized review and case actions

Resident
Authenticated account and complaint tracking

Staff Dashboard (Next.js / TypeScript)
HITL review; case progress; supervised updates and approved retraining control

Resident Portal (Next.js / TypeScript)
Subject and complaint narrative; optional date/time and Purok/Sitio or landmark; case status

Application API (FastAPI / Python)
Authentication / authorization; input validation; routing; audit logging

Text Classification Service (Proposed Deployment)
Privacy-aware NLP → Fitted Word + Character TF-IDF → Versioned Calibrated Linear SVM

Predicted Valid?

Yes: Valid

No: Suspicious / Spam

Human-in-the-Loop Review
Suspicious / Spam only
Verify text label; Accept / Clarify / Reject

Accepted

Clarify / Reject

Accepted-Case Service
Create tracking record; assign officer
Unresolved → Ongoing → Resolved
Escalation, evidence and reconsideration

Review Disposition Record
Pending clarification or documented rejection
Not an active case

Human-Verified Label Records
Narrative version and verified text label
Stored separately from case outcome

status / receipt

approved release

eligible reviewed labels

Complaint and Case Database (PostgreSQL)
Narrative versions, original predictions, disposition, case state, deadlines and immutable audit events

Offline Quality-Controlled Retraining
Check eligibility / duplicates / class coverage; consolidate existing training data + verified feedback; train and recalibrate candidate; validate performance

Deployment approved?

Yes

Versioned Model Registry
NLP configuration + TF-IDF + SVM + calibrator
Existing model retained when validation fails